

SHIVA SOMESH

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EDUCATION

Vellore Institute of Technology (VIT), Chennai

Expected June 2028

B.Tech, Computer Science & Engineering (AI & ML) · CGPA: 8.1/10

Coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, Systems Programming, Machine Learning, Deep Learning, Computer Architecture

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, SQL, HTML/CSS, Go

Frameworks: React, Next.js 14, Node.js, Express.js, Flask, TailwindCSS, Flutter, TensorFlow, PyTorch, scikit-learn

Systems & Infra: Distributed Systems, Concurrency, Multi-threading, Synchronization, Scalable Architecture, System Design, REST APIs, CI/CD, GitHub Actions, Docker, Firebase GCP, Supabase, Oracle Cloud ARM

Tools: Git, OpenCV, YOLO (Ultralytics), CrewAI, LangSmith, Sentry, Postman, Linux, Performance Profiling, Debugging, Agile

EXPERIENCE

Android Club, VIT Chennai — Technical Department Member

Oct 2025–Present

- Recruited via RecipeNest submission — production Flutter app evaluated on architecture, software design, CI/CD pipeline quality, and Firebase integration by the club's technical panel.
- Review peer Android projects for design patterns, concurrency correctness, and performance bottlenecks; deliver structured feedback aligned with scalable mobile architecture principles.
- Co-organize internal workshops on Flutter development, state synchronization, and automated testing — fostering a culture of engineering excellence across club members.

PROJECTS

RecipeNest — Production Android Recipe App ↗

2025–Present

Flutter · Firebase Firestore · Firebase Auth · GitHub Actions CI/CD · Google AI Studio

- Built cross-platform mobile app with Firebase Firestore backend — real-time recipe browsing, full-text search, and concurrent multi-user read/write with synchronized persistent state.
- Architected GitHub Actions CI/CD pipeline with automated APK assembly via `./gradlew assembleDebug`, resolving Node.js 24 constraints; enforced build validation and failure debugging on every commit.
- Audited full architecture via Google AI Studio — identified and resolved state synchronization issues, performance bottlenecks, and scalability gaps through systematic code analysis.

DigitalWill — AI Estate Manager [ARC Night 2026 — CyberTech Winner] ↗

2026

Next.js 14 · TypeScript · Supabase · NextAuth v5 · Kimi K2.5 LLM · Shadcn/UI

- Delivered full-stack system in 24 hrs: OAuth Gmail scanner + LLM risk-scoring + inactivity-triggered beneficiary alert pipeline — won CyberTech domain, Microsoft Innovations Club hackathon.
- Designed distributed module ownership strategy for zero merge-conflict parallel development; implemented Next.js 14 App Router + NextAuth v5 zero-trust auth across all data access paths.

MiroFish — AI Social Simulation Engine ↗

2025

Python · SQLite · WebSocket/SSE · pytest · GitHub Actions

- Built distributed agent-based simulation engine generating synthetic behavioral populations; architected SQLite-backed async task queue with real-time WebSocket/SSE for concurrent agent state management.
- Implemented hallucination gate, abstract memory layer, and opinion drift system; validated end-to-end behavior via pytest suite spanning 7 test files with GitHub Actions CI regression testing.

Skin Cancer Detection System [IEEE Research Paper]

2025

Python · TensorFlow · Flask · OpenCV · YOLO · HAM10000

- Co-authored IEEE paper benchmarking CNN, AlexNet, ResNet-50, EfficientNetB0 on HAM10000 across 7 lesion classes; applied transfer learning and augmentation for reliability and generalization.
- Deployed model behind Flask REST API with async preprocessing; profiled and debugged inference latency under concurrent request load.

ACHIEVEMENTS

- ARC Night 2026 (Microsoft Innovations Club, VIT Chennai) — CyberTech domain winner; built DigitalWill as a 2-person team in 24 hours.
- IEEE Research — Co-authored comparative deep learning paper benchmarking 4 CNN architectures on HAM10000 for skin lesion classification.